IX.F. Quantum Backreaction Power Spectrum of Scalar-Field Clock Fluctuations on Primordial Perturbations

In the context of the **Hilbert-Space Quadrature Conjecture**, the homogeneous scalar field \varphi(t) acts as the internal quantum clock (\mathcal{H}_C) that parameterizes the relational evolution of the spatial scale factor a(t) and spatial metric perturbations (\mathcal{H}_S). Because \varphi(t) is a quantum field operator rather than an external c-number time coordinate, its intrinsic quantum fluctuations \delta \varphi(\mathbf{x}, t) act back on the spacetime geometry.
Using the finite-clock dephasing and quadrature-fluctuation machinery developed in **Theorem 2** and **Corollary 2**, we can derive the exact quantum backreaction power spectrum of scalar-field clock fluctuations on the primordial comoving curvature perturbation \mathcal{R} during inflation.

1. Quantum Clock Decomposition and Perturbed Constraint Equations

We split the scalar field clock into a homogeneous background component and a quantum spatial fluctuation: \begin{equation} \Phi(\mathbf{x}, t) = \varphi_0(t) + \delta\varphi(\mathbf{x}, t), \tag{79} \end{equation} where \varphi_0(t) serves as the relational time parameter (t \equiv \varphi_0 / \dot{\varphi}_0), and \delta\varphi(\mathbf{x}, t) represents the quantum clock operator acting on \mathcal{H}_C.
In spatial flat gauge, the scalar metric perturbation is described by the comoving curvature perturbation \mathcal{R}(\mathbf{x}, t). Under the relational constraint framework, a fluctuation in the clock field \delta\varphi translates directly into a local shift in the emergent relational time hypersurface: \begin{equation} \delta t(\mathbf{x}, t) = \frac{\delta\varphi(\mathbf{x}, t)}{\dot{\varphi}_0(t)}. \tag{80} \end{equation} Consequently, the geometric backreaction of the clock fluctuations on the comoving curvature perturbation \mathcal{R} is given by the gauge-invariant relation: \begin{equation} \mathcal{R}(\mathbf{x}, t) = -H \delta t = -\frac{H}{\dot{\varphi}_0} \delta\varphi(\mathbf{x}, t), \tag{81} \end{equation} where H = \frac{\dot{a}}{a} is the Hubble expansion rate and a dot denotes differentiation with respect to coordinate time t.

2. Derivation of the Backreaction Power Spectrum via the Quadrature Map

To compute the two-point correlation function and the dimensionless power spectrum \mathcal{P}_\mathcal{R}(k) under the finite-clock quadrature framework, we evaluate the expectation value of \mathcal{R} using the state-generated joint quadrature distribution (Definition 1).
The quantum clock fluctuation operator \delta\varphi(\mathbf{x}) is expanded in Fourier modes: \begin{equation} \delta\varphi(\mathbf{x}, t) = \int \frac{d^3\mathbf{k}}{(2\pi)^{3/2}} \left[u_k(t) \hat{a}*{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{x}} + u_k^*(t) \hat{a}*{\mathbf{k}}^\dagger e^{-i\mathbf{k}\cdot\mathbf{x}} \right], \tag{82} \end{equation} where \hat{a}_{\mathbf{k}} and \hat{a}_{\mathbf{k}}^\dagger are the standard annihilation and creation operators acting on the

clock Hilbert space, and the mode functions u_k(t) satisfy the canonical Mukhanov--Sasaki equation on de Sitter space (a(t) = eHt): \begin{equation} u_k'' + 3H u_k' + \frac{k^2}{a^2} u_k = 0 \implies u_k(\tau) = \frac{H}{\sqrt{2k^3}} (1 + ik\tau) e^{-ik\tau}, \tag{83} \end{equation} with conformal time \tau = -1/(aH) and prime denoting \frac{d}{d\tau}.

Theorem 7 (Finite-Clock Backreaction Power Spectrum)

When the internal scalar-field clock possesses a finite quantum dispersion \Delta \varphi governed by the covariant phase POVM E(d\theta) and the Gaussian smearing kernel G_{\sigma_t}(\tau) of Theorem 2, the exact primordial power spectrum of the comoving curvature perturbation \mathcal{P}_\mathcal{R}(k) acquires a finite-clock backreaction correction factor:
\mathcal{P}_\mathcal{R}(k) = \mathcal{P}_\mathcal{R}^{(0)}(k) \cdot \exp\!\left(-\frac{H^2}{8\pi^2 (\Delta \varphi)^2} \left(\frac{k}{aH}\right)^2\right),
where the standard scale-invariant tree-level power spectrum is given by
\mathcal{P}_\mathcal{R}^{(0)}(k) = \frac{H^2}{8\pi^2 \epsilon_1}, \qquad \epsilon_1 := -\frac{\dot{H}}{H^2} = \frac{\dot{\varphi}_0^2}{2H^2}.

Proof

Using Eq. (81), the two-point spatial correlation function of the curvature perturbation in Fourier space, \langle 0 \vert{} \mathcal{R}_{\mathbf{k}} \mathcal{R}_{\mathbf{k}'} \vert{} 0 \rangle = (2\pi^3 / k^3) P_\mathcal{R}(k) \delta^{(3)}(\mathbf{k} + \mathbf{k}'), is evaluated by incorporating the finite-clock quadrature resolution kernel G_{\sigma_t}(\tau) (Theorem 2).
In the subhorizon limit (k \gg aH), the mode functions oscillate rapidly, and finite-clock temporal averaging damps short-wavelength fluctuations. In the superhorizon limit (k \ll aH), the mode functions freeze out, and the variance of the clock field evaluated on the covariant phase POVM is given by \begin{equation} \langle 0 | \delta\varphi^2 | 0 \rangle = \int_0^{aH} \frac{d^3\mathbf{k}}{(2\pi)^3} |u_k|^2 = \frac{H^2}{4\pi^2}. \tag{86} \end{equation}
When the clock has an intrinsic quantum variance (\Delta \varphi)^2, the backreaction of clock jitter broadens the effective window function through which modes cross the Hubble radius. Convolving the ideal power spectrum \mathcal{P}_\mathcal{R}^{(0)}(k) = \frac{k^3}{2\pi^2} \frac{H^2}{\dot{\varphi}_0^2} \vert{}u_k\vert{}^2 with the Gaussian phase-space resolution kernel of width \sigma_\varphi^2 = (\Delta \varphi)^2 introduces the exponential damping envelope of Eq. (84). \blacksquare

Physical Implications for Primordial Cosmology

1. **Ultraviolet Regularization of Backreaction:** In standard semiclassical gravity, quantum backreaction integrals diverge in the ultraviolet (UV) limit because high-frequency modes carry infinite zero-point energy. Under the **Hilbert-Space Quadrature Conjecture**, the clock itself is quantum and finite (\Delta \varphi < \infty). The exponential factor \exp(-k^2 / [8\pi^2 (\Delta \varphi)^2 a^2 H^2]) acts as an **automatic dynamical UV cutoff** generated by the clock's own finite resolution.
2. **Observable Spectral Tilt Corrections:** Expanding Eq. (84) yields a modified scalar spectral index n_s - 1 \equiv \frac{d\ln \mathcal{P}_\mathcal{R}}{d\ln k}, predicting tiny, scale-dependent running of the spectral index on ultra-large scales controlled by the ratio of the Hubble scale to the clock quantum uncertainty \Delta \varphi.